



International Collaboration for the Development of TwinCAT Motion Control Software for Neutron Instruments

Control System Upgrades

Simon Cooper (ISIS)*

Mechatronics Engineer

ISIS Design Division - Motion Control

simon.cooper@stfc.ac.uk

Stephen Cox (ISIS), Dominic Oram (ISIS),

Federico Rojas (ESS), Torsten Bögerhausen (ESS), Paul Barron (ESS),

Frank Suxdorf (JCNS), Harald Kleines (JCNS), Michael Klein (JCNS), Sven Janaschke (JCNS)



Agenda

1. Overview of current motion control systems at ISIS facility
2. Selection of new motion control system
3. Project plan for implementation of new controller
4. The TwinCAT working group
5. Working arrangements and future goals

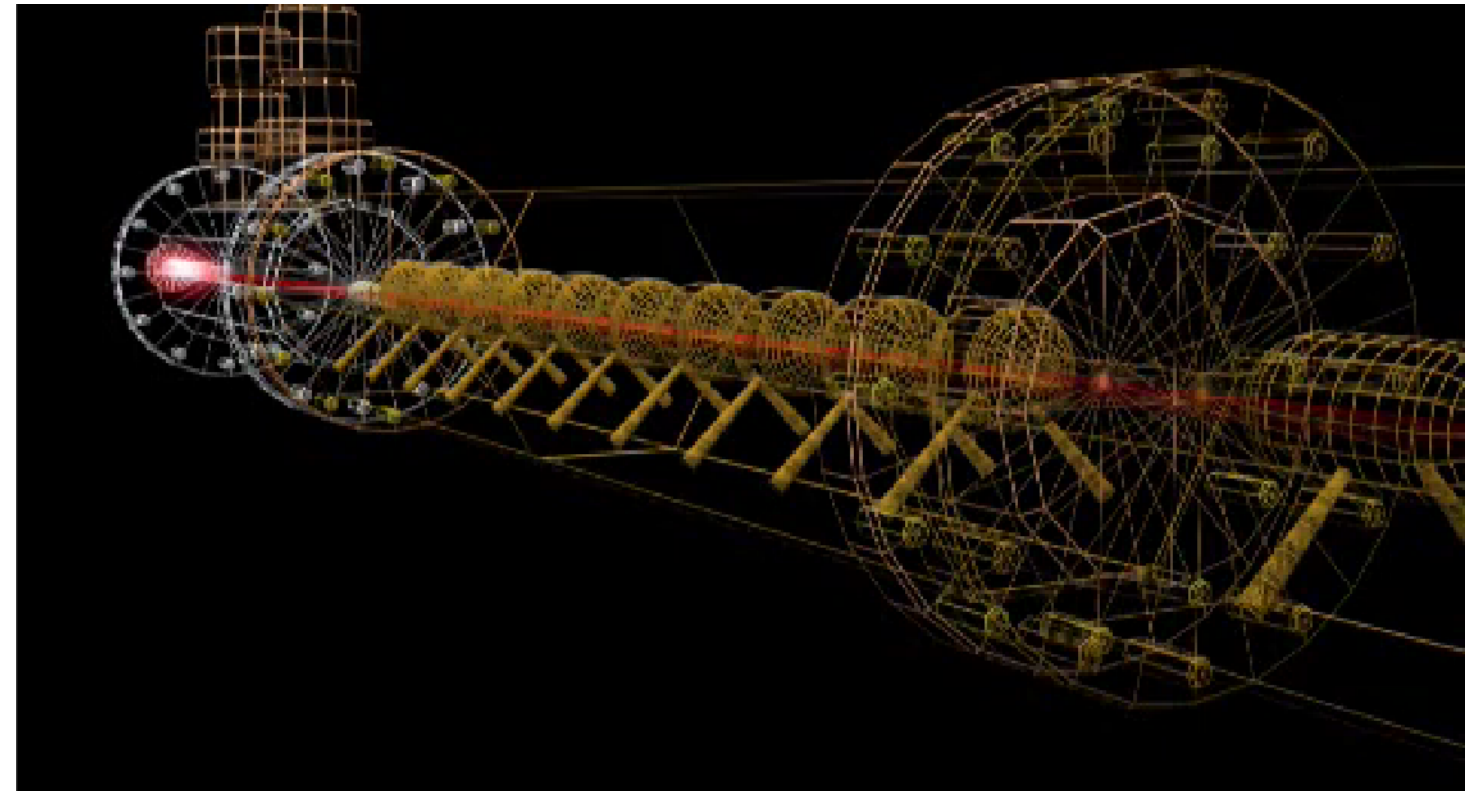


- Neutron and muon pulsed spallation source, located in Oxfordshire, UK
- Two instrument target stations
 - Facility and Target Station 1 opened 1985
 - Target Station 2 opened 2008
- Over 30 operation beamlines
- 350 staff members



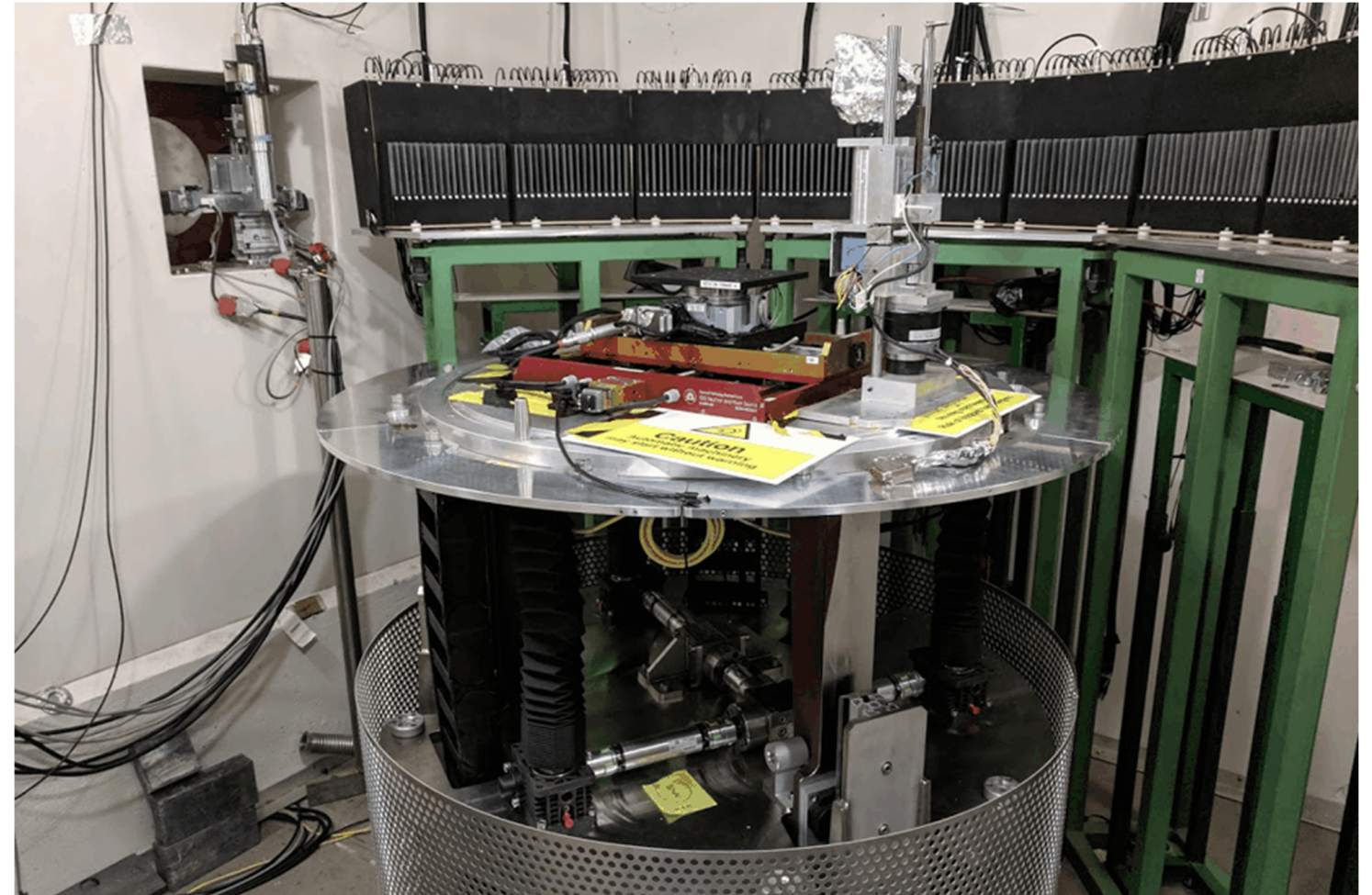
ISIS Neutron and Muon Source

- Neutron and muon pulsed spallation source, located in Oxfordshire, UK
- Two instrument target stations
 - Facility and Target Station 1 opened 1985
 - Target Station 2 opened 2008
- Over 30 operation beamlines
- 350 staff members



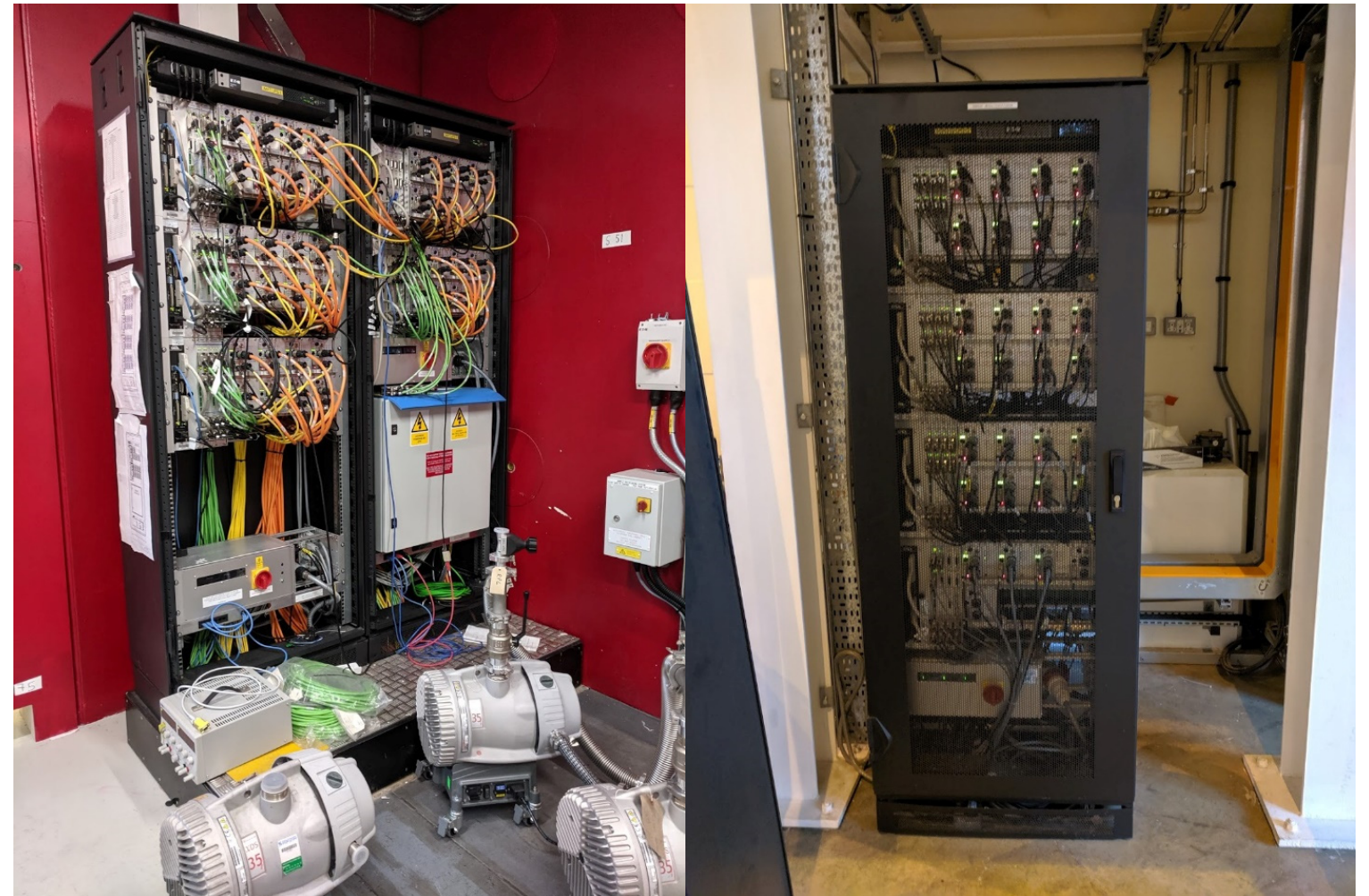
Current Motion Control Systems

- Managed by a combination of:
 - Design engineers
 - Operational support staff
 - Computing support staff
- Standardised across beamlines
 - Accelerators run under different system
- Running over 600 axes of motion
- Typical standard axis consists of:
 - Stepper motor
 - Incremental encoder
 - End-of-Travel limit switches
 - Datum/Home switch
- Typical applications are:
 - Neutron collimating jaw sets
 - Sample positioning systems
 - Sample changers



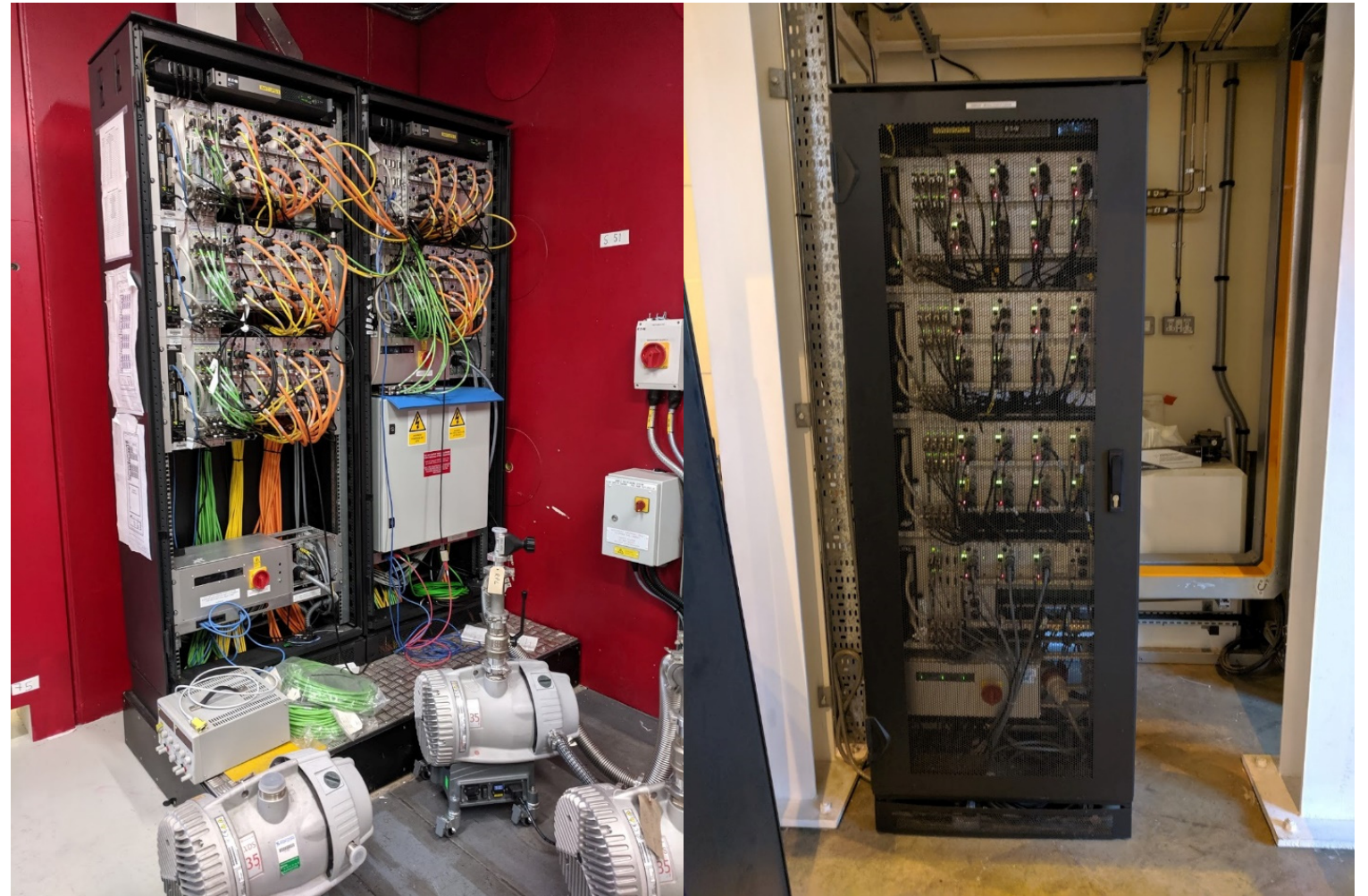
Current Motion Control Systems

- GALIL based system
 - Uses in-house designed and built stepper drive cards
- Very minimal logic located on the controller
 - Generally limited to homing setup values
 - Was done due to resource limitations and attempting to keep all controllers standard
 - Memory/processing power pretty limited (~300 lines of code possible)
- User/scientist interaction through IBEX control layer running EPICS
 - EPICS motor record used for position control
 - Common interface for *all* axes of motion



Current Motion Control Systems

- Poor hardware version control and documentation
 - Unknown what racks run what cards
 - Some incorrectly labelled
- Current setup cannot reliably support anything other than our standard axis arrangement
- Not able to keep up with our evolving science/motion requirements
- Never expected to be a standard or widespread system
 - Facility needed a solution for TS2



Selection of Control System for Upgrade

- Tender to find new system
- Project board of engineers from computing, operations, and design, and scientists to generate system requirements
 - One lead representative from each group
- Tender tailored towards offering the best flexibility
 - Aiming for a family of parts/scalable solution
 - Not “one size fits all”
- Something that could keep up with the constantly evolving motion requirements and requests of our scientists.
- As a team our goal is ensuring that motion is not the blocker holding back better science



Control System Requirements

ESSENTIAL REQUIREMENTS	DESIRABLES
Closed loop control stepper/encoder	Clock synchronisation with external clock (neutron beam pulse)
Synchronisation of at least 9 motors from a master	Position timestamping
Incremental encoder interface	Profile moves
Virtual axes	Table of point moves
Axis position streaming <20ms	Step/direction interface (to retain compatibility with some of our much older hardware)
Absolute encoder and resolver interface	Analog drive interface
EtherCAT interface	Support for hand-held jog box
External hardware triggering from positions	Power down of individual drives
Control of motion controller from external PC	Support external hardware movement triggers
No control cards required in external PC	Digital I/O interface
STO on drive cards	Analog I/O interface



Control System Requirements

ESSENTIAL REQUIREMENTS	DESIRABLES
Closed loop control stepper/encoder	Clock synchronisation with external clock (neutron beam pulse)
Synchronisation of at least 9 motors from a master	Position timestamping
Incremental encoder interface	Profile moves
Virtual axes	Table of point moves
Axis position streaming <20ms	Step/direction interface (to retain compatibility with some of our much older hardware)
Absolute encoder and resolver interface	Analog drive interface
EtherCAT interface	Support for hand-held jog box
External hardware triggering from positions	Power down of individual drives
Control of motion controller from external PC	Support external hardware movement triggers
No control cards required in external PC	Digital I/O interface
STO on drive cards	Analog I/O interface



New Controller Implementation Plan

- Beckhoff selected from tender process
 - Controller uses a software PLC called TwinCAT
- Roll-out broken in to three tranches
 - Tranche 1: Business as usual
 - Development of base code solution in TwinCAT
 - Initial development of operational support capabilities
 - Tranche 2: Meeting current needs
 - Advancing support capabilities; preventative rather than reactive maintenance
 - Delivering solutions to applications we have not previously been able to
 - Tranche 3: Meeting future requirements
 - Development roadmap to keep ahead of the science requirements and improve functionality
- Just completed Tranche 1
 - 2 online systems running with EPICS/IBEX
 - 1 offline systems being commissioned and used for reliability testing



BECKHOFF

TwinCAT®



Science & Technology Facilities Council
ISIS Neutron and Muon Source

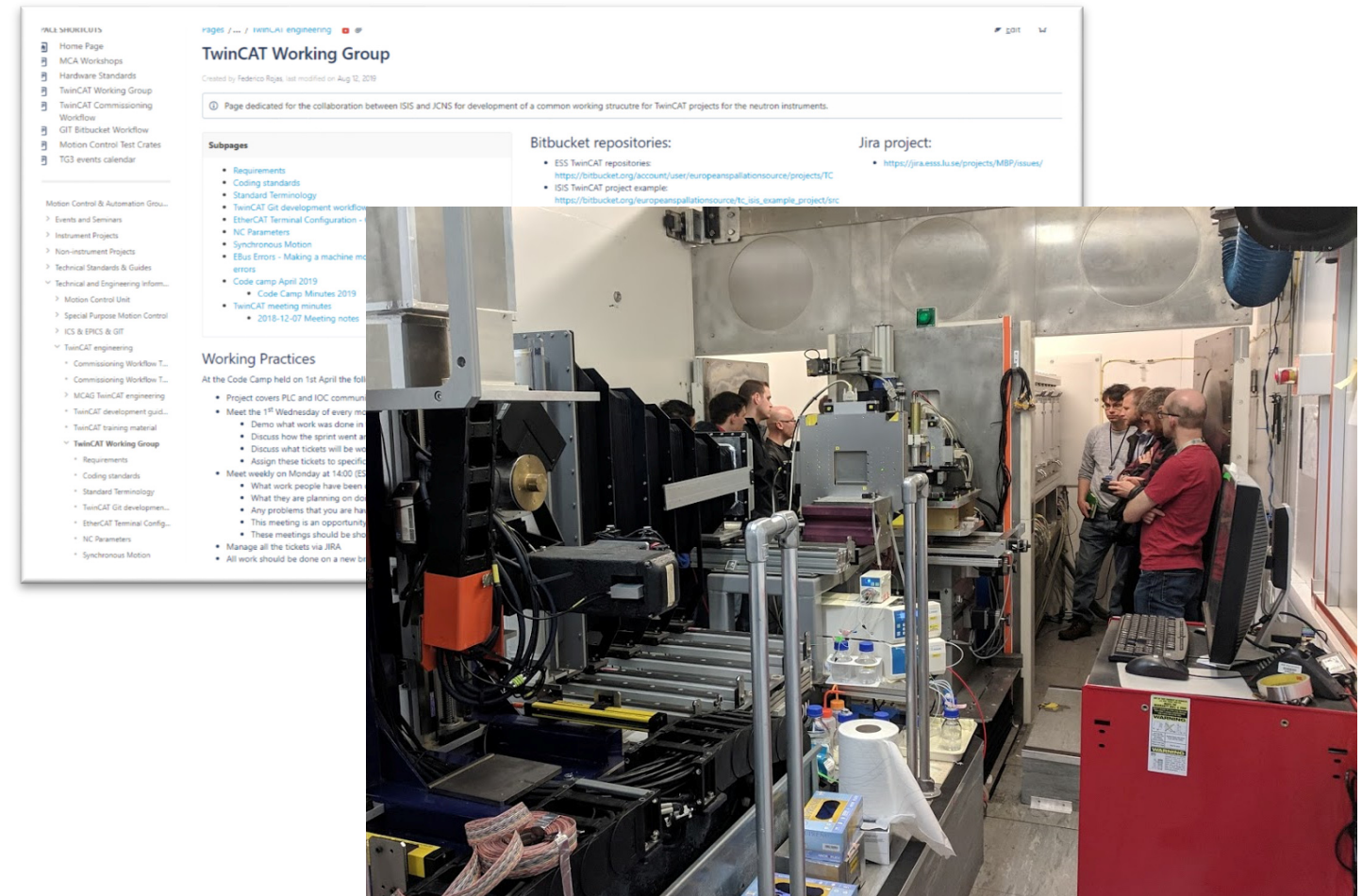
Creation of the TwinCAT Working Group

- Other facilities using TwinCAT for Motion Control
- New neutron research facility being built in Sweden
 - 2023 start of user programme
- In-Kind partnership to deliver instruments for the new ESS facility
 - ISIS is one of 40 European partner institutions who make up the more than 130 collaborating institutions worldwide
- One day TwinCAT event to discuss code solutions
 - Attended by ESS, ISIS, and JCNS
 - Each facility presented the PLC code they are currently using for motion and lessons learned
 - Idea of a collaborative code base proposed
- One week “code camp” arranged at ISIS



Working Arrangements and Future

- Code camp agenda:
 - Grouped requirements generation
 - Tours of instruments from reflectometry scientists
 - Presentations and workshops
- Outputs:
 - Confluence pages used for documentation
 - Jira Kanban style board for software tickets
- Weekly short video conference
- Monthly longer video conference
- Ticket system for prioritising and delivering work
 - Decide features as a group
 - New branch for development
 - Review by at least one other facility



Working Arrangements and Future

- The arrangement is working
 - Single base code being developed
 - Axis synchronisation implemented
 - Controller-based solutions to applications we've previously had to solve at EPICS layer
- Not been easy
 - Every facility is different
 - Diverse group of engineers
 - None of us are software engineers
 - Different delivery timelines
- But has been worth the effort
 - More people
 - Different ideas to problems
 - Feature requests from scientists may be already covered as facilities will use the system differently
 - More systems on more beamlines means more likely to be robust





Thank you for listening

If you are a facility using Beckhoff and TwinCAT for motion control systems and would like to be involved in this project please get in touch.

Contact details:

Simon Cooper MEng (Hons) MIET

FS Tech (TÜV Rheinland, # 1536/ 18, Machinery)

Mechatronics Engineer

ISIS Design Division - Motion Control

simon.cooper@stfc.ac.uk

Links:

ISIS Neutron and Muon Source: <https://www.isis.stfc.ac.uk>

European Spallation Source: <https://europeanspallationsource.se/>

Jülich Centre for Neutron Science: <https://www.fz-juelich.de/jcns>